

Dean Kuhn

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Summary

Computer science graduate (WGU, May 2026) focused on data engineering and pipeline development. Built and deployed end-to-end data systems: a full-stack analytics platform serving longitudinal growth data for 22,000+ artists, and an ML demand-forecasting pipeline with automated A/B evaluation and live Power BI/Streamlit reporting. Seeking a junior data engineering, analytics engineering, or data analysis role.

Education

B.S. Computer Science · Western Governors University · May 2026

M.S. Computer Systems · Western Governors University · Expected January 2027

Technical Portfolio

Music Growth Pipeline · 2026

Tech: Python · PostgreSQL (Neon) · dbt Core · Next.js · TypeScript · Power BI · GitHub Actions

- Built and deployed a full-stack analytics platform (**music.deanslist.dev**, Next.js 15) on top of a weekly-automated pipeline snapshotting 22,000+ artists from Last.fm — plus a live Power BI report reading from the same warehouse.
- Found that median 13-week listener growth falls monotonically with starting audience size across 22,201 artists (2.67% smallest quintile → 1.71% largest, zero reversals), with the smallest artists' upside tail over 4x larger at P90.
- Modeled the transformation layer in dbt (staging + mart models, LAG window functions) feeding both the web app's serving layer and the Power BI report.

KitchenSync Food Forecasting System · 2026

Tech: Python · FastAPI · PostgreSQL (Neon) · LightGBM · Streamlit · Power BI · AWS EC2 (dbt/Snowflake — prior architecture)

- Built an end-to-end demand forecasting system for a simulated 12-store retail kitchen chain: async POS ingestion, nightly feature rebuild, LightGBM forecasting, and a Streamlit dashboard driving live production queues — fully deployed on AWS EC2 with automated nightly runs.
- Weather-bucketed A/B testing against a naive baseline isolated the model's real edge: **+2.0pp service level on extreme-temperature days vs. +0.7pp on neutral days**, proving the model exploits a signal the baseline structurally can't access — including an honest negative result on precipitation days where the baseline briefly wins on service level.
- Diagnosed two silent data-quality bugs via root-cause debugging: a day-of-week convention mismatch that zeroed **2.37M training rows**, and a conditional-mean bias inflating predictions 3–4x at low-traffic stores.

VRP-TW Genetic Algorithm Routing Optimizer (Python) — a genetic algorithm solving package-to-truck assignment and delivery sequencing simultaneously via a single sentinel-chromosome fitness function, avoiding the two-pass approach most implementations use.

Technical Skills

Languages: Python, SQL

Data & Storage: PostgreSQL, Snowflake, dbt Core (dbt-postgres, dbt-snowflake), pandas, PyArrow, Parquet

Visualization: Power BI, Streamlit, Plotly

Cloud & Infra: AWS EC2, GitHub Actions, Docker, Neon (cloud Postgres), systemd, Git, Linux/WSL

Ingestion & APIs: FastAPI, REST APIs, asyncio/httpx, Playwright (headless scraping)

ML & Modeling: LightGBM, scikit-learn

Concepts: ETL/ELT pipeline design, medallion architecture, data warehouse modeling, dbt staging/mart patterns, ML inference serving

Exposure: DuckDB, dbt-duckdb, MotherDuck, Cloudflare R2 (object storage), pgvector/RAG

Work Experience

Guest Services & Inventory Coordinator · Kwik Trip · 2021 – Present

- Tracked inventory and managed ingredient purchasing for kitchen operations in a high-volume food production environment (the same system KitchenSync models), balancing stock daily to avoid both shortages and excess waste.
- Ran end-of-shift cash reconciliation and register audits across high daily transaction volume; caught and resolved discrepancies before close.
- First-line troubleshooting for POS terminal failures, network issues, and delivery problems during live shifts with no dedicated IT support on site.